■ Exploratory Data Analysis (EDA) - Student Dataset

Task Objective:
To perform Exploratory Data Analysis on a student dataset and extract meaningful insights using statistical and visualization techniques.

Tools & Technologies Used:
- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Dataset Description:
The dataset contains student-related information such as gender, study hours, attendance, and scores in various subjects.

Steps Performed:
1. Imported required libraries
2. Loaded dataset using Pandas
3. Performed data inspection using info() and describe()
4. Checked for missing values
5. Conducted univariate analysis using histograms
6. Identified outliers using boxplots
7. Performed bivariate analysis using scatter plots
8. Generated pairplots for multi-variable relationships
9. Created correlation heatmap for feature relationships
10. Derived insights from the data

Key Insights:
- Students with higher study hours tend to score better
- Attendance positively impacts performance
- Strong correlation observed between reading and writing scores
- Minimal performance difference based on gender

Conclusion:
EDA helps in understanding patterns, detecting anomalies, and preparing data for further machine learning tasks.

Task Done By:
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